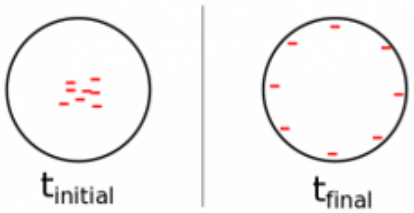
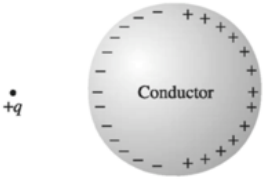
Chapter 37, Electric Field, Conductors and Applications
Calculus Physics II

Charge migration in a conductor describes the movement of free charges in response to an electric field until a state of electrostatic equilibrium is reached. Unlike insulators, where charges are immobile, conductors have freely moving electrons that can redistribute themselves quickly to neutralize any internal electric field.

For a net charge placed on a conductor at $t_{initial}$, the charge will spread out over the surface of the conductor due to their natural repulsion shown in t_{final}. There will be no excess charge inside the conductor, all charge will migrate to the surface due to the repulsion of the like charge.

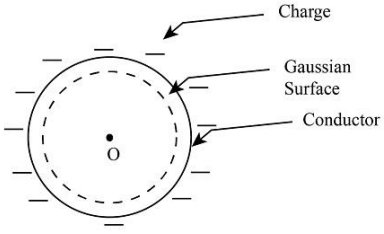


If a net charge is brought near a conductor, the charge on that conductor will redistribute to neutralizes the external electric field created by the net charge. Shown is a net positive charge near a neutral conductor, the negative charge on the conductor is attracted to the net positive charge and the positive charge on the conductor is repelled by the net positive charge.



<u>Electric field inside a Conductor</u> - Inside a conductor in electrostatic equilibrium, the electric field is zero. This occurs because free electrons within the conductor move in response to any applied electric field, redistributing themselves until they cancel out the field internally. As a result, any excess charge resides only on the surface of the conductor, and the interior becomes electrically neutral.

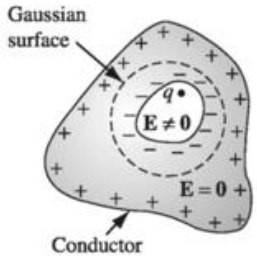
- At equilibrium, under electrostatic conditions, any excess charge resides on the surface of a conductor because it is repelled in all directions till it reaches the surface. Under these conditions, the electric field is zero at any point within a conducting material. There is no charge within the Gaussian surface shown therefore no electric field.
- If there were an electric field in a conductor, then charge would experience a force due to this electric field and move therefore once the charge is in equilibrium the electric field inside a conductor must be zero.



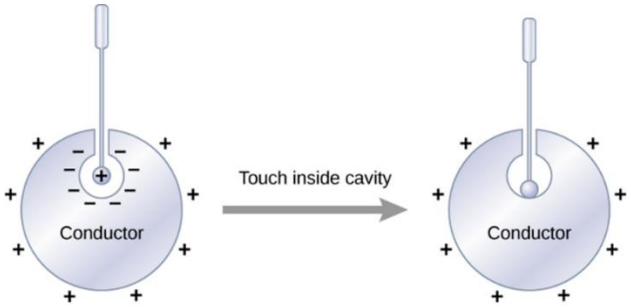
If a net charge is placed inside an empty cavity in a conductor, this charge will create an electric field in the empty space of the cavity and induce an equal but opposite type of charge on the inside surface of this empty cavity. In the conductor itself, a Gaussian surface would enclose both the point charge and the induced surface charge, this would cancel the net charge in the Gaussian surface therefore the electric field in the conductor must be zero.

The induced charge on the inside of the conductor creates a net charge on the outside of the conductor with equal but opposite type.

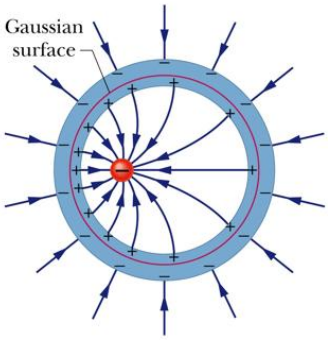
A Gaussian surface around the whole conductor would include the charge in the empty cavity, therefore there is a electric field outside the conductor.



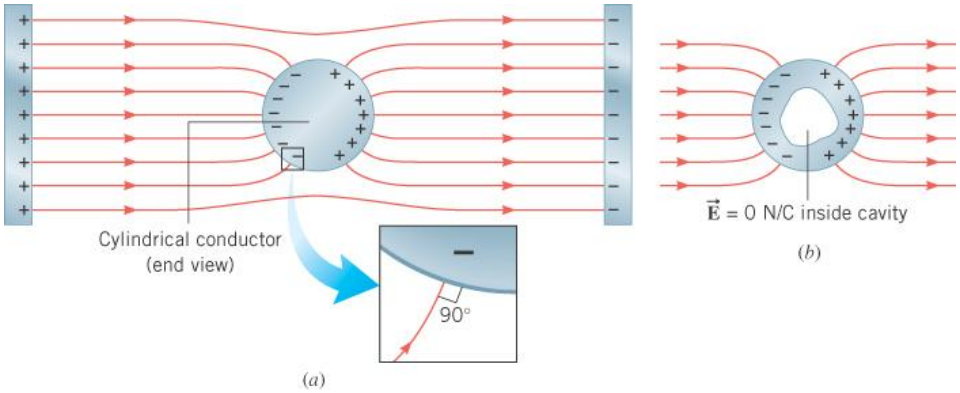
This concept is shown in the diagram below, a charged rod is placed in an empty cavity of the conductor without touching it, this charge induces the same charge on the conductor's surface. If the rod comes in contact with the conductor and the charge is transferred to the conductor, the excess charge will spread out over the outside surface of the conductor.



A net charge placed at any location inside a cavity inside the conductor (as long as it does not touch the surface of the conductor) will create a uniform charge on the outside surface of the conductor because the charge on the outside experiences a mutual repulsion. Also, the electric field lines must enter or exit a conductor perpendicular to its surface which means that the charge must be uniformly distributed on the outside of the conductor.

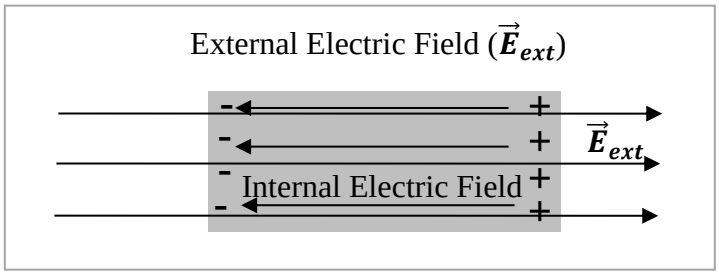


<u>A Conductor in an Electric field</u> - A conductor in an electric field shields its inside volume from any electric fields outside the conductor and the field line terminates on the conducting surface at right angles.

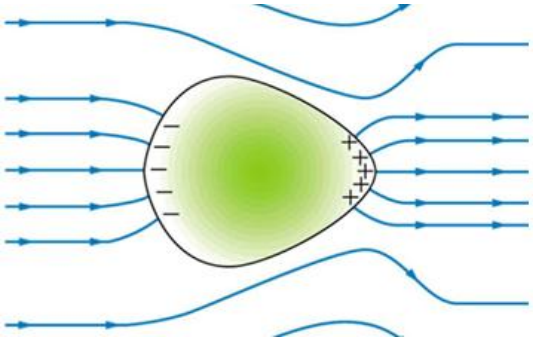


- ❖ The electric field inside a conductor is zero because if there was an electric field it would move the charge in a way that would cancel the field.
- ❖ The external electric field is canceled by the induced internal electric field

When a conductor is placed in an electric field the positive charge moves along the external electric field lines in the direction of the field and the same about of negative charge moves in the opposite direction. This charge separation creates an internal electric field (induced field) that is opposite in direction but equal in magnitude to the external electric field so that the total net field inside the conductor is zero.



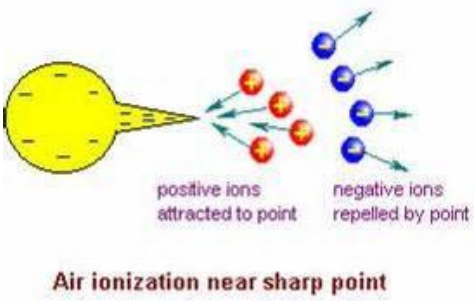
For a conductor in an electric field, the field lines must enter and exit perpendicular to the conductor surface therefore the external field lines are distorted by the conductor's presences.



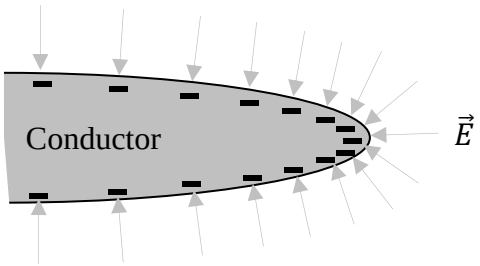
Conductors in Electrostatic Equilibrium

1) Any excess charge that is placed on a conductor moves to the surface and stays there.
2) The electric field inside of a conductor is zero anywhere inside the conductor.
3) The electric field strength outside the conductor is largest at sharp corners (small radius).
4) The electric field lines entering or exiting the surface of a conductor are always perpendicular to the surface of the conductor.
5) The entire conductor is at the same electric potential throughout its volume.

Electric field at corners of conductors - For an irregular shaped conductor the excess charge is more concentrated at regions of smaller radius of curvatures.



Air ionization near sharp point



- Because of the large electric fields generated at the tips of conductors the air around the tip can become ionized (electrons stripped from the atoms leaving free electrons and positive charged atoms). This creates charged particles around the conductor's tip.

Surface/Area Charge Density (σ - sigma) – For a conductor all charge is distributed on the surface of the conductor therefore a conductor with a net charge of Q and a surface area of A, the surface charge density is given as

$$\sigma = \frac{Q}{A} \quad units\ are\ \frac{C}{m^2}$$

For the surface of a conducting sphere the area $A = 4\pi r^2$ therefore

$$\sigma_{Sphere} = \frac{Q}{A} = \frac{Q}{4\pi r^2} \quad - \quad \text{for a conducting sphere}$$

- The larger the radius of the conducting sphere the smaller the surface charge density on that sphere
- For a uniform conducting sphere, the charge density is uniform but if the object has curved surfaces that have different radius, then the surface charge density on the object is not uniform and varies by the surface curvature.

$$\sigma \text{ is proportional to } \frac{1}{r^2}$$

- The smaller the radius the larger the surface charge density.

The electric field generated by a net charge on a conductor is equal to the total charge on that conductor which is The surface charge density

$$Q_{in} = \sigma A_{sphere}$$

$$|\vec{E}| = \frac{Q_{in}}{(A_{Gausian})\varepsilon_o} = \frac{\sigma A_{sphere}}{(A_{Gausian})\varepsilon_o}$$

➢ The larger the surface charge density the larger the electric field strength

Example: The same charge of q = 5 nC is placed on two conducting spheres with different radii, Sphere 1 has a radius of r_1 = 2 cm and Sphere 2 has a radius of r_2 = 20 cm.

A) Determine the surface charge density on each sphere.

$$\sigma_1 = \frac{Q}{4\pi r^2} = 9.95 \text{ x } 10^{-7}\ C/m^2$$

$$\sigma_2 = \frac{Q}{4\pi r^2} = 9.95 \text{ x } 10^{-9}\ C/m^2$$

B) Determine the electric field strength at the surface each sphere, assume a Gaussian surface at the radius of each sphere. $A_{sphere} = A_{Gaussian}$ $\varepsilon_o = 8.85\ x\ 10^{-12}\ C^2/Nm^2$

$$|\vec{E}| = \frac{Q_{in}}{(A_{Gaussian})\varepsilon_o} = \frac{\sigma A_{sphere}}{(A_{Gausian})\varepsilon_o} = \frac{\sigma}{\varepsilon_o}$$

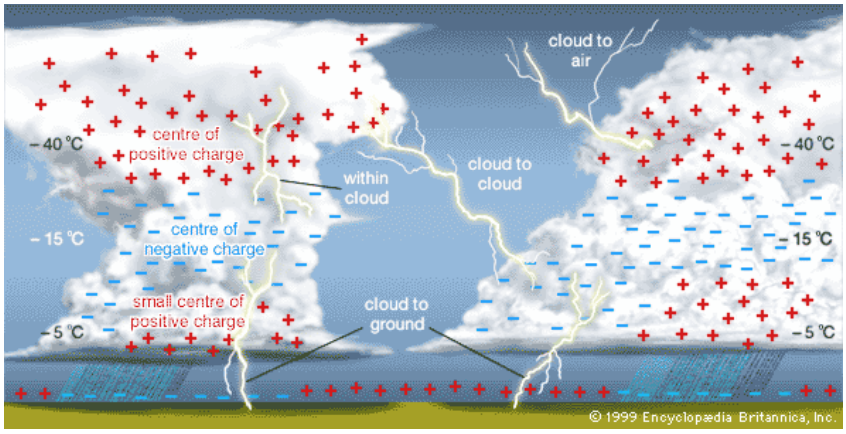
$$Shere\ 1\ |\vec{E}| = \frac{\sigma_1}{\varepsilon_o} = 1.12 \text{ x } 10^5\ N/C$$

$$Shere\ 2\ |\vec{E}| = \frac{\sigma_2}{\varepsilon_o} = 1.12 \text{ x } 10^3\ N/C$$

❖ The smaller sphere has an electric field strength that is 100 times larger than the larger radius sphere.

<u>Example</u>: **Lighting, Charge and the Electric Field** – lightning is produced when the water molecules and droplets in the clouds rub together as the move up and down in the cloud (and from earth to the cloud), this rubbing strips electrons off the water molecules and this charge is transported to different areas of the cloud creating a charge separation in the cloud and with the ground below.

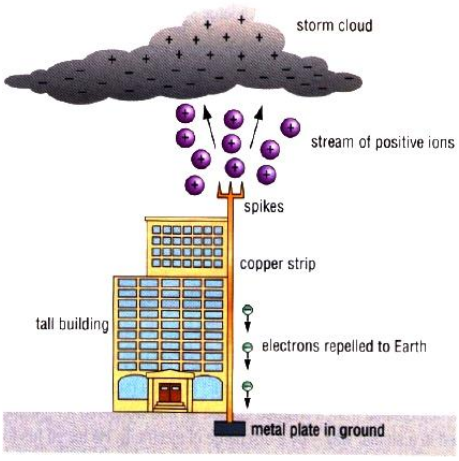
When the charge on the cloud is large enough and a large electric field is generated the air molecules are ionized creating a path for the electrons to flow, this is usually to the ground, but it can be within the cloud.



<u>Lightning rod</u> – a lightning rod works because it is a conductor that is connected to ground, charge can flow between the rod and ground. The tip of the rod is sharp so that when the rod is charged it creates a very large electric field that is capable of ionizing the air around it. When a negatively charged cloud moves near the lightning rod it induces positive charge to move up the rod from the ground which ionizes the air around the rod, the positive ions in the air move towards the negatively charged cloud thereby keeping the air from being ionized by the cloud and not letting the charge flow from the cloud.

- The Lighting rod also protects the building so if lightning does strike the charge will flow down the rod to ground and not though the building.





Coronal Discharge – When air or a gas is ionized due to the high electric fields produced at the tip of a charged conductor, the electrons and ions eventually recombine, this recombination releases energy in the form of light, the color of the light that is emitted by this recombination depends on the type of gas that was involved.

Coronal Discharge is seen as a soft glow around the tip of the conductor.

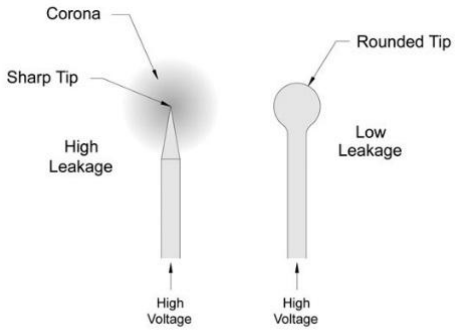
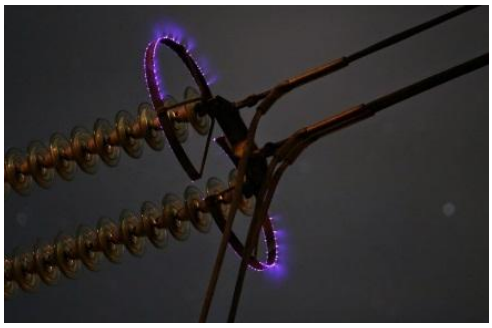


Figure 1





Examples of Coronal Discharge

- Sharp edges on high voltage power lines.
- Glowing around tips of mast of ships (St. Elmo's fire) due to charged clouds.

Electric Fields and Power Lines – The picture below is of fluorescent light tubes stuck in the ground below high voltage power lines. The electric field generated by the power lines is strong enough to ionize the gas in the tubes lighting them up as if they were plugged in the lighting fixture.



Faraday cage - A Faraday's cage demonstrates the fact that the electric field is zero inside a conductor. When an external Electric or Electromagnetic Field (EM Field) hits the conductor, the charges within the cage's material redistribute themselves to creating an opposing field, effectively canceling the external field inside the cage. For static electric fields, the charges move to the outside surface, while for changing electromagnetic waves, they induce eddy currents that generate opposing magnetic fields, which also block the wave from passing through.

The large machine (Tesla Coil) generates a very large electric field that ionizes the air (sparks you see) sending electric charge to the metal cage (a conductor) but inside the car or cage a man is standing and is not being affected by the electric field because he is surrounded by the wire metal conductor. Conducting packages are used to shield sensitive electronics from electric fields.







- Faraday cages are effective at blocking static electric fields and electromagnetic waves, such as radio waves and microwaves. They are used to protect against lightning strikes, as the electrical current flows around the outside of the cage to the ground.

- Faraday cages cannot block static or slowly varying magnetic fields. They also do not block visible light, as visible light is a part of the electromagnetic spectrum, but the cage's mesh is typically large enough to allow visible light to pass through.

Static electricity buildup can damage sensitive electronics; Electrostatic Discharge (ESD) is the sudden energy release, which grounding prevents by safely dissipating charge via wrist straps, mats, or special flooring, creating an ESD Protected Area where personnel and equipment are bonded to a common earth ground, ensuring safety and component reliability.



